

LAB 4 - DJANGO PROJECT

All project files combined into ONE text file.

This file contains the complete source code for the project.

To run it as a Django project, the files should be separated according to the FILE: headings into their original folders.

```
=====
====
FILE: README.md
=====
=====
```

Lab 4 - Django Models, QuerySets & Media Files

This project covers the Lab 4 requirements:

- Models
- Fields and Attributes
- QuerySet Methods
- Media file upload

1. Install

```
```bash
```

```
python -m venv venv
```

```
Windows:
```

```
venv\Scripts\activate
```

```
Linux/macOS:
```

```
source venv/bin/activate
```

```
pip install -r requirements.txt
```

```
```
```

2. Create the database

```
``bash  
python manage.py makemigrations  
python manage.py migrate  
...
```

3. Create an admin account

```
``bash  
python manage.py createsuperuser  
...
```

4. Run

```
``bash  
python manage.py runserver  
...
```

Open:

- Home: <http://127.0.0.1:8000/>
- Add product: <http://127.0.0.1:8000/products/add/>
- Admin: <http://127.0.0.1:8000/admin/>

QuerySet examples included

The project demonstrates:

- all()
- filter()
- exclude()
- get()
- order_by()
- count()
- exists()
- first()
- last()

Media

Products have an optional ImageField. Uploaded images are stored in `media/products/`.

```
=====
=====
```

FILE: lab4_project/__init__.py

```
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```

```
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```

FILE: lab4_project/settings.py

```
=====
=====
```

```
from pathlib import Path
```

```
BASE_DIR = Path(__file__).resolve().parent.parent
```

```
SECRET_KEY = "django-insecure-lab4-demo-key-change-this"
```

```
DEBUG = True
```

```
ALLOWED_HOSTS = []
```

```
INSTALLED_APPS = [
```

```
    "django.contrib.admin",
```

```
    "django.contrib.auth",
```

```
    "django.contrib.contenttypes",
```

```
    "django.contrib.sessions",
```

```
    "django.contrib.messages",
```

```
"django.contrib.staticfiles",
"products",
]
```

```
MIDDLEWARE = [
    "django.middleware.security.SecurityMiddleware",
    "django.contrib.sessions.middleware.SessionMiddleware",
    "django.middleware.common.CommonMiddleware",
    "django.middleware.csrf.CsrfViewMiddleware",
    "django.contrib.auth.middleware.AuthenticationMiddleware",
    "django.contrib.messages.middleware.MessageMiddleware",
    "django.middleware.clickjacking.XFrameOptionsMiddleware",
]
```

```
ROOT_URLCONF = "lab4_project.urls"
```

```
TEMPLATES = [
    {
        "BACKEND": "django.template.backends.django.DjangoTemplates",
        "DIRS": [BASE_DIR / "templates"],
        "APP_DIRS": True,
        "OPTIONS": {
            "context_processors": [
                "django.template.context_processors.request",
                "django.contrib.auth.context_processors.auth",
                "django.contrib.messages.context_processors.messages",
            ],
        },
    ],
]
```

```
WSGI_APPLICATION = "lab4_project.wsgi.application"
```

```
DATABASES = {
    "default": {
        "ENGINE": "django.db.backends.sqlite3",
        "NAME": BASE_DIR / "db.sqlite3",
    }
}
```

```
AUTH_PASSWORD_VALIDATORS = []
```

```
LANGUAGE_CODE = "en-us"
```

```
TIME_ZONE = "Africa/Cairo"
```

```
USE_I18N = True
```

```
USE_TZ = True
```

```
STATIC_URL = "static/"
```

```
STATICFILES_DIRS = [BASE_DIR / "static"]
```

```
MEDIA_URL = "/media/"
```

```
MEDIA_ROOT = BASE_DIR / "media"
```

```
DEFAULT_AUTO_FIELD = "django.db.models.BigAutoField"
```

```
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```

```
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```

```
FILE: lab4_project/urls.py
```

```
=====
```

```
=====
```

```
from django.conf import settings
```

```
from django.conf.urls.static import static
```

```
from django.contrib import admin
```

```
from django.urls import include, path
```

```
urlpatterns = [  
    path("admin/", admin.site.urls),  
    path("", include("products.urls")),  
]
```

```
if settings.DEBUG:
```

```
    urlpatterns += static(settings.MEDIA_URL, document_root=settings.MEDIA_  
ROOT)
```

```
=====
```

```
FILE: lab4_project/wsgi.py
```

```
=====
```

```
import os
```

```
from django.core.wsgi import get_wsgi_application
```

```
os.environ.setdefault("DJANGO_SETTINGS_MODULE", "lab4_project.settings")  
application = get_wsgi_application()
```

```
=====
```

```
FILE: manage.py
```

```
=====
```

```
#!/usr/bin/env python
```

```
import os
```

```
import sys
```

```
def main():
    os.environ.setdefault("DJANGO_SETTINGS_MODULE", "lab4_project.settings")
    from django.core.management import execute_from_command_line
    execute_from_command_line(sys.argv)

if __name__ == "__main__":
    main()
```

=====

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FILE: media/.gitkeep

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FILE: products/__init__.py

=====

=====

=====

=====

FILE: products/admin.py

=====

=====

```
from django.contrib import admin
```

```
from .models import Product
```

```
@admin.register(Product)
```

```
class ProductAdmin(admin.ModelAdmin):
```

```
    list_display = ("name", "category", "price", "stock", "is_available", "created_at")
```

```
    list_filter = ("category", "is_available")
```

```
    search_fields = ("name", "description", "category")
```

```
    ordering = ("-created_at",)
```

```
=====
```

```
====
```

```
FILE: products/apps.py
```

```
=====
```

```
====
```

```
from django.apps import AppConfig
```

```
class ProductsConfig(AppConfig):
```

```
    default_auto_field = "django.db.models.BigAutoField"
```

```
    name = "products"
```

```
=====
```

```
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```

```
FILE: products/forms.py
```

```
=====
```

```
====
```

```
from django import forms
```

```
from .models import Product
```

```
class ProductForm(forms.ModelForm):
    class Meta:
        model = Product
        fields = [
            "name", "description", "price", "category",
            "stock", "image", "is_available"
        ]
        widgets = {
            "description": forms.Textarea(attrs={"rows": 4}),
        }
```

```
=====
=====
```

FILE: products/models.py

```
=====
=====
```

```
from django.db import models
from django.urls import reverse
```

```
class Product(models.Model):
    # Fields and attributes
    name = models.CharField(max_length=100)
    description = models.TextField(blank=True)
    price = models.DecimalField(max_digits=10, decimal_places=2)
    category = models.CharField(max_length=50, default="General")
    stock = models.PositiveIntegerField(default=0)
    image = models.ImageField(upload_to="products/", blank=True, null=True)
    is_available = models.BooleanField(default=True)
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)
```

```

class Meta:
    ordering = ["-created_at"]
    verbose_name = "Product"
    verbose_name_plural = "Products"

def __str__(self):
    return self.name

def get_absolute_url(self):
    return reverse("product_detail", args=[self.pk])

@property
def in_stock(self):
    return self.stock > 0 and self.is_available

```

=====

====

FILE: products/urls.py

=====

====

```

from django.urls import path
from . import views

```

```

urlpatterns = [
    path("", views.home, name="home"),
    path("products/add/", views.add_product, name="add_product"),
    path("products/<int:pk>/", views.product_detail, name="product_detail"),
]

```

=====

====

FILE: products/views.py

=====

====

```
from django.shortcuts import get_object_or_404, redirect, render
```

```
from .forms import ProductForm
```

```
from .models import Product
```

```
def home(request):
```

```
    # QuerySet methods: filter + order_by
```

```
    products = Product.objects.filter(is_available=True).order_by("-created_at")
```

```
    # More QuerySet examples for the lab
```

```
    context = {
```

```
        "products": products,
```

```
        "product_count": Product.objects.count(),
```

```
        "available_count": Product.objects.filter(is_available=True).count(),
```

```
        "has_products": Product.objects.exists(),
```

```
    }
```

```
    return render(request, "products/home.html", context)
```

```
def product_detail(request, pk):
```

```
    product = get_object_or_404(Product, pk=pk)
```

```
    return render(request, "products/detail.html", {"product": product})
```

```
def add_product(request):
```

```
    if request.method == "POST":
```

```
        form = ProductForm(request.POST, request.FILES)
```

```
        if form.is_valid():
```

```
            product = form.save()
```

```
            return redirect(product.get_absolute_url())
```

```
    else:
```

```
form = ProductForm()
```

```
return render(request, "products/add.html", {"form": form})
```

```
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```

```
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```

```
FILE: requirements.txt
```

```
=====
```

```
=====
```

```
Django>=5.0,<6.0
```

```
Pillow>=10.0
```

```
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```

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```

```
FILE: static/.gitkeep
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```
FILE: templates/base.html
```

```
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```

```
=====
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>{% block title %}Lab 4{% endblock %}</title>
<style>
    body { font-family: Arial, sans-serif; max-width: 1000px; margin: 0 auto; padding: 20px; background:#f5f5f5; }
    nav { background:#222; padding:15px; margin-bottom:25px; border-radius:8px; }
    nav a { color:white; text-decoration:none; margin-right:20px; }
    .card { background:white; padding:20px; margin:15px 0; border-radius:10px; box-shadow:0 2px 8px #ddd; }
    .grid { display:grid; grid-template-columns:repeat(auto-fit,minmax(250px,1fr)); gap:15px; }
    img { max-width:100%; max-height:220px; border-radius:8px; }
    .btn { display:inline-block; padding:10px 15px; background:#222; color:white; text-decoration:none; border-radius:6px; border:0; cursor:pointer; }
    input, textarea, select { width:100%; padding:10px; margin:6px 0 14px 0; box-sizing:border-box; }
    label { font-weight:bold; }
</style>
</head>
<body>
<nav>
    <a href="{% url 'home' %}">Lab 4 Products</a>
    <a href="{% url 'add_product' %}">Add Product</a>
    <a href="/admin/">Admin</a>
</nav>
{% block content %}{% endblock %}
</body>
</html>
```

```
=====
=====
```

FILE: templates/products/add.html

=====

=====

```
{% extends "base.html" %}
{% block title %}Add Product{% endblock %}
{% block content %}
<div class="card">
    <h1>Add Product</h1>
    <form method="post" enctype="multipart/form-data">
        {% csrf_token %}
        {{ form.as_p }}
        <button class="btn" type="submit">Save Product</button>
    </form>
</div>
{% endblock %}
```

=====

=====

FILE: templates/products/detail.html

=====

=====

```
{% extends "base.html" %}
{% block title %}{{ product.name }}{% endblock %}
{% block content %}
<div class="card">
    {% if product.image %}
        
    {% endif %}
    <h1>{{ product.name }}</h1>
    <p>{{ product.description }}</p>
</div>
```

```
<p><strong>Category:</strong> {{ product.category }}</p>
<p><strong>Price:</strong> ${{ product.price }}</p>
<p><strong>Stock:</strong> {{ product.stock }}</p>
<p><strong>Available:</strong> {{ product.is_available|yesno:"Yes,No" }}</p>
<p><strong>In stock property:</strong> {{ product.in_stock|yesno:"Yes,No" }}
</p>
<a class="btn" href="{% url 'home' %}">Back</a>
</div>
{% endblock %}
```

```
=====
=====
```

FILE: templates/products/home.html

```
=====
=====
```

```
{% extends "base.html" %}
{% block title %}Products - Lab 4{% endblock %}
{% block content %}
<h1>Lab 4 - Products</h1>
<p>Total products: <strong>{{ product_count }}</strong> | Available: <strong>{{
available_count }}</strong></p>

{% if has_products %}
<div class="grid">
{% for product in products %}
<div class="card">
    {% if product.image %}
        
    {% endif %}
    <h2>{{ product.name }}</h2>
    <p><strong>Category:</strong> {{ product.category }}</p>
```

<p>Price: \${{ product.price }}</p>

<p>Stock: {{ product.stock }}</p>

View details

</div>

{% endfor %}

</div>

{% else %}

<div class="card">

<p>No products yet. Add one from the button above.</p>

</div>

{% endif %}

{% endblock %}